Growth of functions solutions

03-E01 Prove bounds from quantifiers

For n≥3, $f/n^2 = 5 - 9/n + 2/n^2 \geq 2$. For n≥1, −9n+2≤0, giving f≤5n². Thus c_1=2,c_2=5,n_0=3 work. If f≤cn eventually, combining with f≥2n² gives 2n≤c. Choose an integer $n > \max(n_0, 3, c/2)$, a contradiction for any candidate c. At n=3, the check is 18≤20≤45, but the inequalities establish the general result. Hence f∈Θ(n²) and f∉O(n).

03-E02 Loose upper bounds and little-o

The ratio f/g is 1 on even n and 1/n on odd n. Since f≤g, O holds. An Ω constant c>0 would require 1/n≥c for every large odd n, which fails; hence Θ also fails. Little-o fails with c=1/2 on even n. Little-ω fails with c=2 on even n. A loose O bound need not be little-o. For $h = n^2(2 + (-1)^n)$, ratios alternate 1 and 3. The bounds g≤h≤3g establish Θ(g) despite the nonexistent ratio limit.

03-E03 General properties and cancellation

Nonnegativity gives max(f,g)≤f+g≤2max(f,g), with fixed constants 1 and 2. For O transitivity, f≤ag and g≤bh beyond a common threshold imply f≤abh. For little-o transitivity, given any c>0, use constant 1 for the first relation and c for the second. Beyond their common threshold, f<g<ch.

The difference need not be Θ(h): choose f=g=h=n, giving zero. Even a nonnegative difference need not have a positive lower bound. A negative difference also falls outside the chapter's nonnegative O definition unless an absolute-value bound is specified.

03-E04 Input quantifiers

The extrema are $T_{min} = n$ and $T_{max} = n^2$. Thus their tight orders are Θ(n) and Θ(n²); common bounds are O(n²) and Ω(n). If all inputs had a common Θ(g) bound, the n-cost inputs would force g=O(n), while the n²-cost inputs would force n²=O(g). Transitivity would imply n²=O(n), a contradiction. A worst-case lower bound applies to hard inputs, not every easy input.

03-E05 Integer splits and negative mod

The interval has length 11. Its midpoint is $q = 7 + \lfloor (17 - 7)/2 \rfloor = 12$, giving lengths 6 on 7..12 and 5 on 13..17. Floor(−11/4)=−3, ceiling=−2, and −11 mod 4=−11−4(−3)=1.

Growth of functions solutions (continued)

For an integer k, $k \le \lfloor \lfloor x/a \rfloor / b \rfloor$ is equivalent to $kb \le \lfloor x/a \rfloor$, and, since kb is integral, to kab≤x. This is equivalent to $k \le \lfloor x/(ab) \rfloor$. Both expressions therefore have the same greatest integer. A truncated quotient −2 instead gives remainder −3, different from the specified nonnegative mod.

03-E06 Logarithm and exponent traps

The listed order is increasing. The first ratio tends to zero by fixed-polylogarithm domination. Next,

$$\frac{n^{0.1}}{n \lg n} = \frac{1}{n^{0.9} \lg n} \to 0, \qquad \frac{n \lg n}{n^2} = \frac{\lg n}{n} \to 0.$$

The binomial lower bound for 3^n gives $n^2/3^n \to 0$. The factorial final-factor bound gives $3^n/n! \to 0$. Also $2^{\lg^2 n} = n^{\lg n}$, whose ratio to every fixed n^k is $n^{\lg n - k} \to \infty$, so it is not polynomially bounded.

We have $3^{n+2} = 9 \cdot 3^n$, hence $\Theta(3^n)$. In contrast, $3^{2n}/3^n = 3^n$ is unbounded, so the second function is neither $O(3^n)$ nor $\Theta(3^n)$. Multiplicative constants in an exponent cannot be discarded like constants from a log-base change.

03-E07 General factorial bounds

Use $(n/2)^{\lfloor n/2 \rfloor} \le n! \le n^n$. Taking logarithms gives Θ(n log n). Dividing each of the first floor(n/2) factors by n bounds their ratios by 1/2 and the remaining ratios by 1, so $n!/n^n \le 2^{-\lfloor n/2 \rfloor} \to 0$. For fixed a>0,

$$\ln(n!/a^n) \ge \lfloor n/2 \rfloor \ln(n/2) - n \ln a \to \infty,$$

proving $\omega(a^n)$.

At n=5, the Stirling main term is $S = \sqrt{10\pi}(5/e)^5 \approx 118.019168$. Factors e^(1/61) and e^(1/60) give approximately 119.96985<120<120.00264. Relative error n!/S−1 is Θ(1/n); absolute error n!−S is S • Θ(1/n), which does not tend to zero. The three growth results above did not depend on Stirling.

03-E08 Iteration and stopping

The chain is 256→8→3→lg 3≈1.584963→lg(lg 3)≈.664449, taking four iterations. The second iterate is 3; the square of the first logarithm is 64.

Growth of functions solutions (continued)

```
LOG-STAR(n)
    k = 0
    while n > 1
        n = lg(n)
        k = k + 1
    return k
```

A logarithm is taken only when n>1, so its result is positive. A value in (1,2] goes to at most 1 next. For n>2, lg n<n. If iterates remained at least 2 forever, the decreasing sequence would have a limit L≥2. Continuity would force L=lg L, contradicting lg L<L on that range. Hence termination holds in the real-number model.

Floating-point overflow can produce infinity, with lg(infinity)=infinity and no termination. Reject nonfinite inputs, require n>0, and define rounding and stopping policies. Finite precision does not exactly represent arbitrary real inputs.

03-E09 Fibonacci closed form and representation cost

The proposed G has $G_0=0, G_1=1$. Both root-power sequences obey the same second-order recurrence because their roots satisfy $z^2=z+1$. Therefore G does too; induction with matching bases proves G=F. The sequence 0,1,1,2,3,5,8,13,21 gives $F_8=21$. Numerically, $\phi^8/\sqrt{5}\approx 21.009519$ and $\psi^8/\sqrt{5}\approx .009519$, whose difference is 21.

The absolute error from $\phi^n/\sqrt{5}$ is less than 1/2. For sufficiently large n, fixed positive multiples bracket F_n, proving $\Theta(\phi^n)$. Thus $\lg F_n=\Theta(n)$. Exact output requires $\Theta(n)$ bits; with word size $\Theta(\log n)$, writing it needs at least $\Omega(n/\log n)$ words, precluding $O(1)$. Approximation errors in ϕ and exponentiation can grow, so final rounding alone does not guarantee exact results for arbitrary n.

03-E10 Anonymous functions and uniform constants

The equality means that every $h\in\Theta(n)$ makes $6n^2+h(n)$ a member of $\Theta(n^2)$. It does not say every $\Theta(n^2)$ function has that form. If $h(i)\le ci$ for $i\ge i_0$, let the finite prefix sum be K. Then

$$\sum_{i=1}^{n} h(i) \le K + c\sum_{i=1}^{n} i \le K + cn(n+1)/2 = O(n^2).$$

One c works for the whole tail. The family $h_n(i)=ni$ has two parameters; its apparent O(i) constant n depends on the outer upper limit. It cannot be treated as independent of n after summation. The actual sum is $n^2(n+1)/2=\Theta(n^3)$.

Growth of functions solutions (continued)

Verification code appendix: growth_checks.py

```
"""Finite arithmetic checks, not substitutes for asymptotic proofs."""
from fractions import Fraction as F
from math import log2,sqrt,pi,exp,factorial,ceil,floor
def logstar(x):
 k=0
 while x>1:x=log2(x);k+=1
 return k
if __name__=='__main__':
 for n in range(4,1001):assert n*n<=3*n*n-8*n+5<=3*n*n
 for n in range(3,1001):assert 2*n*n<=5*n*n-9*n+2<=5*n*n
 for a in range(1,8):
  for b in range(1,8):
   for q in range(201):
    x=F(q,10);assert floor(F(floor(x/a),b))==floor(x/(a*b))
    assert ceil(F(ceil(x/a),b))==ceil(x/(a*b))
 assert -11%4==1 and logstar(256)==4 and logstar(65536)==4
 phi=(1+sqrt(5))/2;psi=(1-sqrt(5))/2;f=[0,1]
 for n in range(2,31):f.append(f[-1]+f[-2])
 for n,v in enumerate(f):assert round((phi**n-psi**n)/sqrt(5))==v
 for n in [5,6]:
  s=sqrt(2*pi*n)*(n/exp(1))**n
  lo=s*exp(1/(12*n+1));hi=s*exp(1/(12*n))
  assert lo<factorial(n)<hi
  print('Stirling',n,'main',s,'bounds',lo,hi)
 print('PASS finite: explicit polynomial constants, 9849 rational nested floors/ceilings,
log-star and Fibonacci n<=30')
```

Lin Cing https://github.com/xinhi6716/computer-science-self-study

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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